

Haoqiang (Murray) Kang

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RESEARCH PROFILE

I develop generalist foundation models that **perceive, reason, generate, and act** across digital and physical environments. My work spans three directions: (1) **LLM/VLM reasoning and agent post-training**, including RL algorithms and environments for (latent) reasoning and coding agents; (2) **multimodal generative models** for unified understanding and generation, including multimodal reasoning and image/video generation; and (3) **embodied foundation models**, including VLAs, world models, world action models, and scalable environments.

EDUCATION

• University of California San Diego (UCSD)

Ph.D. Candidate in Computer Science; Advisor: Lianhui Qin

La Jolla, CA

• University of Washington (UW)

B.S. in Computer Science; Advisor: Luke Zettlemoyer

Seattle, WA

PUBLICATIONS

8+ first-author papers accepted to or under review at top-tier AI/ML conferences, spanning latent reasoning, RL agents, multimodal learning, and embodied AI.

- [1] Haoqiang Kang, Yizhe Zhang, Nikki Lijing Kuang, Yian Ma, Lianhui Qin (2026). **Uni-LaDiR: Latent Diffusion Unifies Multimodal Reasoning**. *Under Review*.
- [2] Haoqiang Kang, Yiming Zhang, Yiyang Guo, Chuying Li, Jianzhi Shen, Tianruo Rose Xu, Xiaokang Ye, Lianhui Qin (2026). **DeliveryGym: An RL Environment for Long-Horizon Embodied Agent Planning with Adaptive Curriculum**. *Under Review*.
- [3] Haoqiang Kang, Yinpeng Chen, Luyang Liu, Jesper Sparre Andersen, Abhijit Ogale, Baochen Sun, Lichan Hong, Ed H. Chi (2026). **Scaffolding Minds: Optimizing Latent Visual Target Representations for Multimodal Reasoning**. *Under Review*.
- [4] Guancheng Tu, Xiangjun Fu, Suhao Yu, Yao Tang, Haoqiang Kang, Lianhui Qin, Yizhe Zhang, Jiatao Gu (2026). **Latent Reasoning with Normalizing Flows**. *Under Review*.
- [5] Haoqiang Kang, Xiaokang Ye, Yuhan Liu, Siddhant Hitesh Mantri, Lingjun Mao, James Fleming, Drishti Regmi, Lianhui Qin (2026). **SimWorld Studio: Automatic Environment Generation with Evolving Coding Agent for Embodied Agent Learning**. *NeurIPS 2026*.
- [6] Haoqiang Kang, Yizhe Zhang, Nikki Lijing Kuang, Yi-An Ma, Lianhui Qin (2026). **LaDi-RL: Latent Diffusion Reasoning Prevents Entropy Collapse in Reinforcement Learning**. *Under Review*.
- [7] Fangxu Yu, Xingang Guo, Lingzhi Yuan, Haoqiang Kang, Hongyu Zhao, Lianhui Qin, Furong Huang, Bin Hu, Tianyi Zhou (2026). **TSRBench: A Comprehensive Multi-task Multi-modal Time Series Reasoning Benchmark for Generalist Models**. *ICML 2026*.
- [8] Lanxiang Hu, Abhilash Shankarampeta, Yixin Huang, Zilin Dai, Haoyang Yu, Yujie Zhao, Haoqiang Kang, Daniel Zhao, Tajana Rosing, Hao Zhang (2025). **Benchmarking Scientific Understanding and Reasoning for Video Generation using VideoScience-Bench**. *ECCV 2026*.
- [9] **PAN: A World Model for General, Interactable, and Long-Horizon World Simulation**.
- [10] Haoqiang Kang, Yizhe Zhang, Nikki Lijing Kuang, Nicklas Majamaki, Navdeep Jaitly, Yi-An Ma, Lianhui Qin (2025). **LaDiR: Latent Diffusion Enhances LLMs for Text Reasoning**. *ICLR 2026*.
- [11] Haoqiang Kang, Enna Sachdeva, Piyush Gupta, Sangjae Bae, Kwonjoon Lee (2024). **GFlowVLM: Enhancing Multi-step Reasoning in Vision-Language Models with Generative Flow Networks**. *CVPR 2025*.
- [12] Fangxu Yu*, Lai Jiang*, Haoqiang Kang*, Shibo Hao, Lianhui Qin (2024). **Flow of Reasoning: Training LLMs for Divergent Problem Solving with Minimal Examples**. *ICML 2025*.
- [13] Qianqian Xie, Weiguang Han, Xiao Zhang, ..., Haoqiang Kang, ... (2024). **FinBen: A Holistic Financial Benchmark for Large Language Models**. *NeurIPS Datasets and Benchmarks Track*.
- [14] Haoqiang Kang, Terra Blevins, Luke Zettlemoyer (2024). **Comparing Hallucination Detection Metrics for Multilingual Generation**.
- [15] Shijie Han, Haoqiang Kang, Bo Jin, Xiao-Yang Liu, Steve Y. Yang (2024). **XBRL Agent: Leveraging Large Language Models for Financial Report Analysis**. *ACM ICAIF 2024*.
- [16] Haoqiang Kang*, Terra Blevins*, Luke Zettlemoyer (2024). **Translate to Disambiguate: Zero-shot Multilingual Word Sense Disambiguation with Pretrained Language Models**. *EACL 2024 Main Conference, Oral Presentation*.

- [17] Haoqiang Kang, Juntong Ni, Huaxiu Yao (2023). **EVER: Mitigating Hallucination in Large Language Models through Real-Time Verification and Rectification.**
- [18] Haoqiang Kang, Xiao-Yang Liu (2023). **Deficiency of Large Language Models in Finance: An Empirical Examination of Hallucination.** *NeurIPS ICBINB Workshop.*

EXPERIENCE

- **Google DeepMind** 06/2026 – 09/2026
Student Researcher. Host: Abhijit Ogale and Priyank Jaini. Mountain View, CA
Collaboration with Gemini Robotics team.
 - Conducted research on **motion-tracking representation for world modeling**, capturing task-relevant object and robot dynamics for **embodied foundation models**.
 - Used motion tracking as a shared interface between a **VLM-based System 2 planner** and a **VLA-based System 1 executor**, enabling hierarchical planning and closed-loop execution.
 - Outperformed standard text- and image-based interfaces by **10%** and **7%**, respectively, on robotic manipulation.
- **Google DeepMind** 06/2025 – 09/2025
Student Researcher. Host: Yinpeng Chen Seattle, WA
 - Conducted research on **latent visual reasoning** for Vision-Language Models (VLMs).
 - Developed training methods that integrate System 2-like reasoning images into System 1-like latent tokens and then applied **RL training** to improve reasoning performance.
 - Achieved a **+19%** performance gain over baselines on visual reasoning benchmarks, and resulted in the paper [Scaffolding Minds](#).
- **Honda Research Institute, USA** 05/2024 - 09/2024
Research Intern San Jose, CA
 - Conducted research to enhance the multi-turn reasoning capabilities of Vision-Language Models (VLMs).
 - Developed post-training methods leveraging GFlowNets to improve the model performance on embodied planning tasks.
 - Achieved a **+10%+** performance gain over baselines, and resulted in a CVPR paper [GFlowVLM](#).
- **Columbia University** 08/2023 - 05/2024
Research Intern Remote
 - Investigated hallucination issues in Large Language Models for financial applications.
 - Implemented LLM-based agents to process and analyze XBRL files in the finance domain.
- **Z-Lab in University of Washington** 09/2022 - 05/2024
Research Assistant Seattle
 - Conducted research on multilingual word sense disambiguation using Pretrained Language Models.
 - Evaluated and analyzed multilingual hallucination in LLMs.

SKILLS AND SERVICE

Research: LLM/VLM Reasoning, Reinforcement Learning, Post-Training, Agentic AI, Embodied Foundation Models, VLA Models, World Models, Simulation and Evaluation

Programming: Python, Java

Peer Review: ICLR, NeurIPS, ICML, ECCV, CVPR, ACL, EMNLP